

AI CONCEPTS REPORT

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Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines and computers to perform tasks that normally require human intelligence. AI can learn from data, recognize patterns, make decisions, and solve problems. Today, AI is used in many fields such as healthcare, education, banking, transportation, and entertainment. Examples of AI include chatbots, voice assistants, recommendation systems, and self-driving cars. AI helps improve efficiency and reduces human effort in many tasks.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

Artificial Intelligence (AI) is a broad field that focuses on creating intelligent machines capable of performing human-like tasks.

Machine Learning (ML) is a subset of AI that allows computers to learn from data without being explicitly programmed. It improves performance through experience and training data.

Generative AI is a type of AI that can create new content such as text, images, videos, and audio. Examples include ChatGPT and image generation tools.

Agentic AI is an advanced form of AI that can make decisions, plan actions, and perform tasks independently to achieve specific goals. It can work with minimal human intervention.

Q3. Explain any three open-source AI models available on Hugging Face

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BERT: BERT (Bidirectional Encoder Representations from Transformers) is a popular open-source language model developed by Google. It is used for text classification, question answering, and language understanding tasks.

GPT-2: GPT-2 is an AI model developed by OpenAI. It can generate human-like text and is widely used for content generation, chatbots, and writing assistance.

Stable Diffusion: Stable Diffusion is an image generation model that creates high-quality images from text descriptions. It is widely used for digital art and creative design projects.

These models are available on Hugging Face and can be used by developers for various AI applications.

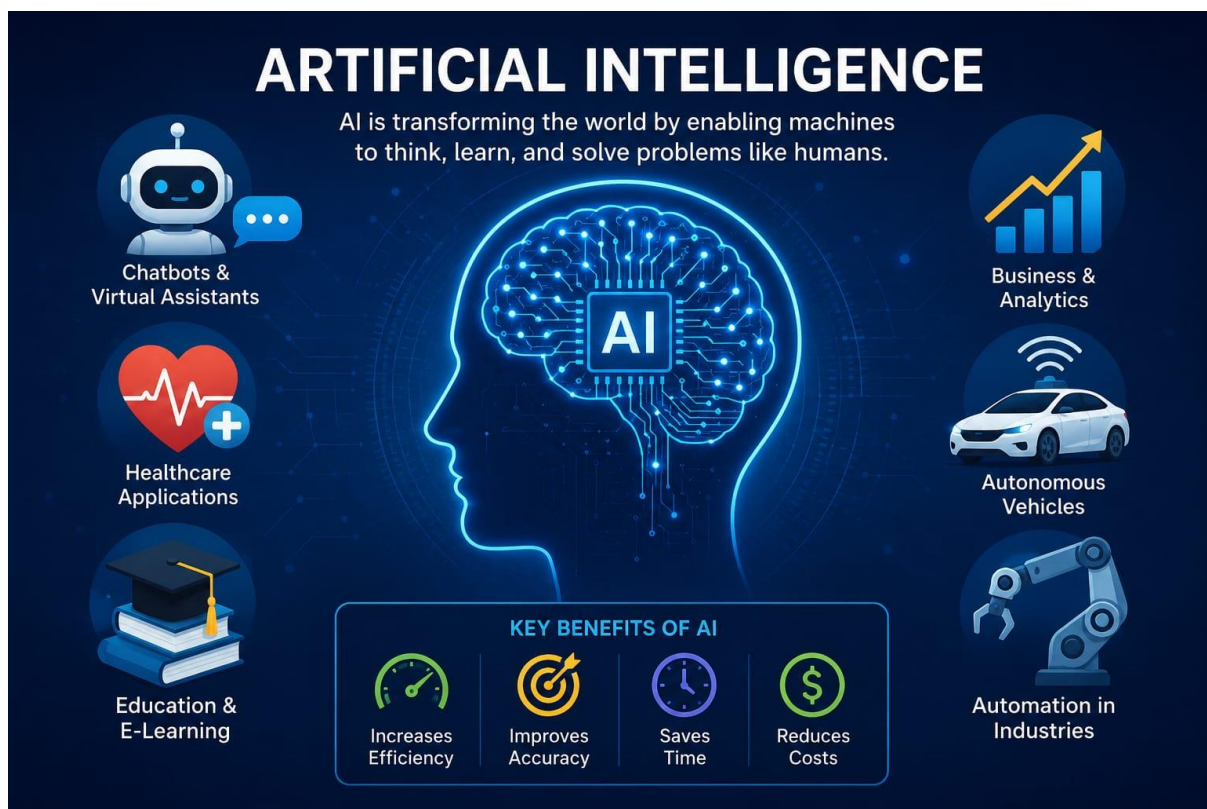


FIGURE 1:ARTIFICIAL INTELLIGENCE APPLICATIONS AND BENEFITS

Q4. Difference between Traditional Programming and AI-Assisted Development

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Traditional programming relies on predefined rules and instructions written by developers. The computer follows these instructions exactly as provided.

AI-assisted development uses AI tools to help developers write code, find errors, generate suggestions, and automate repetitive tasks. It improves productivity and reduces development time.

In traditional programming, developers manually write most of the code. In AI-assisted development, AI tools can generate code snippets and provide intelligent recommendations. This makes software development faster and more efficient.

Q5. Explain AI Hallucinations with one example

AI hallucination occurs when an AI system generates information that is incorrect, misleading, or completely fabricated while presenting it as factual. This happens because AI models predict responses based on patterns in data rather than actual understanding.

For example, if a user asks an AI chatbot for information about a book that does not exist, the AI may generate a fake title, author name, and publication details. Although the response may sound convincing, the information is not real.

To reduce AI hallucinations, users should verify important information using reliable sources and cross-check facts whenever possible.